Junsu Kim

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Research Interests

Computer Architecture, Systems for ML & ML for Systems

Education

University of British Columbia, Vancouver, BC, Canada Sep. 2025 - Current
Ph.D. in Electrical and Computer Engineering (Advisor: Prof. Prashant Nair)
Cumulative GPA: 4.0/4.0

Korea University, Seoul, Korea Sep. 2023 - Aug. 2025
M.S. in Electrical Engineering (Advisor: Prof. Yunho Oh)
Cumulative GPA: 4.0/4.0

Hanyang University, Seoul, Korea Mar. 2014 - Feb. 2021
B.S. in Electronic Engineering (Advisor: Prof. Ki-Seok Chung)
Cumulative GPA: 3.81/4.0 (Summa Cum Laude)

Publications

Conference [C] and Journal [J] Papers (* denotes co-first author)

[C6] Jeonghyun Woo*, **Junsu Kim***, Aamer Jaleel, and Prashant Nair, "SLoaded Dices: Solving the Non-Selection Problem for Scalable Probabilistic RowHammer Defense", The 53rd International Symposium on Computer Architecture (ISCA), 2026.

[J2] **Junsu Kim**, Jaebeom Jeon, Jaeyong Park, and Sangun Choi, Minseong Gil, Seokin Hong, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh "Memory Oversubscription-Aware Tensor Migration Scheduling for GPU Unified Storage Architecture" The IEEE Computer Architecture Letters (CAL), 2025.

[C5] **Junsu Kim**, and Suhyun Kim, "Salient Frequency-aware Exemplar Compression for Resource-constrained Online Continual Learning", The 39th Annual AAAI Conference on Artificial Intelligence (AAAI), 2025.

[C4] Seondeok Kim*, Sangun Choi*, Jaebeom Jeon, **Junsu Kim**, Minseong Gil, Jaehyeok Ryu, and Yunho Oh, "Kubism: Disassembling and Reassembling K-Means Clustering for Mobile Heterogeneous Platforms", The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[C3] Dongwon Yang, Jaebeom Jeon, Minseong Gil, **Junsu Kim**, Seondeok Kim, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "SSFFT: Energy-Efficient Selective Scaling for Fast Fourier Transform in Embedded GPUs", The 26th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems (LCTES), 2025.

[J1] Minseong Gil, Jaebeom Jeon, **Junsu Kim**, Sangun Choi, Gunjae Koo, Myung Kuk Yoon, and Yunho Oh, "TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs", The IEEE Embedded Systems Letters (ESL), 2024.

[C2] Jaebeom Jeon, Minsung Gil, **Junsu Kim**, Jaeyoung Park, Gunjae Koo, Myung-Kuk Yoon, and Yunho Oh. "VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing". The 53rd International Conference on Parallel Processing (ICPP), 2024

[C1] Kwangrae Kim, Jeonghyun Woo, **Junsu Kim**, and Ki-Seok Chung. "HammerFilter: Robust Protection and Low Hardware Overhead Method for RowHammer". The 39th IEEE International Conference on Computer Design (ICCD), 2021

Preprints (Project Names Only)

[P1] Co-author, "Hardware Supports for Enabling Arbitrary Numeric Format on GPUs"

Work Experience

d-Matrix, Santa Clara, CA, United States Jun. 2026 - Sep. 2026
ML Research Intern Mentors: To be determined

University of British Columbia, Vancouver, BC, Canada Sep. 2025 - Current
Research Assistant at Systems and Architectures (STAR) Lab Advisor: Prof. Prashant Nair

Korea University, Seoul, Korea Sep. 2023 - Aug. 2025
Research Assistant at Computer Architecture and System Software Lab (ComSys) Advisor: Prof. Yunho Oh

Korea Institute of Science and Technology, Seoul, Korea May. 2022 - Aug. 2023
Research Assistant at Korea Data Science Team (KDST) Supervisor: Dr. Suhyun Kim

Hanyang University, Seoul, Korea Dec. 2019 - Mar. 2020, Aug. 2020 - Nov. 2020
Research Assistant at Embedded System on Chip Laboratory (ESOC Lab) Advisor: Prof. Ki-Seok Chung
Research Assistant at Computer Architecture and System SW Lab (CASS Lab) Advisor: Prof. Yongjun Park

Teaching Experience

University of British Columbia, Vancouver, BC, Canada Fall 2025
Teaching Assistant for Computer Architecture (CPEN 411)

Korea University, Seoul, Korea Spring 2024, Fall 2024
Teaching Assistant for Computer Architecture

School for the Blind, Chuncheon, Korea Mar. 2017 - Feb. 2019
Assistant Teacher (Alternative Military Service)

Honors and Awards

Faculty of Applied Science Graduate Award, University of British Columbia (UBC) → $7200 CAD Nov. 2025

AAAI Student Travel Grant → ∼ $1600 USD Mar. 2025

Korea University Graduate School Scholarship → Half Tuition (∼ $10000 CAD) Spring 2024, Fall 2024

Korea University Graduate School Scholarship for Outstanding New Student → Half Tuition (∼ $5000 CAD) Fall 2023

Hanyang University Scholarship → Full Tuition (∼ $20000 CAD) 2014 - 2019

Research Projects

Scalable Speculative Decoding with Hierarchical Verification
Advisor: Prof. Prashant Nair, University of British Columbia Mar. 2026 - Current
- ◇ To be updated

Accuracy-preserving Tiered Key-Value Cache Management for Large Language Model
Advisor: Prof. Prashant Nair, University of British Columbia Sep. 2025 - Current
- ◇ To be updated

Solving Non-selection Problem for Probabilistic Rowhammer Defence
Advisor: Prof. Prashant Nair, University of British Columbia Nov. 2025 - Mar. 2026
- ◇ To be updated

Memory Oversubscription-aware Tensor Migration Scheduling for GPU Unified Storage Architecture [CAL]
Advisor: Prof. Yunho Oh, Korea University Feb. 2024 - Sep. 2024
- ◇ Analyzed page faults due to memory oversubscription stalled AI workloads when expanding GPU memory with SSD using UVM
- ◇ Proposed a tensor migration scheduling algorithm considering GPU memory oversubscription for GPU unified storage architecture
- ◇ Achieved the averaged speedup by 12.9% compared to G10, which was presented at MICRO 2023
- ◇ Contributions: 1st author, motivation study, idea, implementation, experiment, paper write-up

Hardware Supports for Enabling Arbitrary Numeric Format on GPUs
Advisor: Prof. Yunho Oh, Korea University May. 2024 - Nov. 2024
- ◇ Observed GPU supports a limited set of numeric formats, wasting register files when processing arbitrary numeric formats
- ◇ Employed bitslice representation, which transposes the data elements, packing arbitrary numeric formats without register wastage.
- ◇ Proposed Bitslice Vector multiplier and adder, constructing a tree structure to replace a multiplication-adder tree in a Tensor core
- ◇ Contributions: co-author, motivation study, idea, implementation, paper write-up

A Behavioral Analysis of CXL Memory Systems
Advisor: Prof. Yunho Oh, Korea University Sep. 2023 - Sep. 2024
Collaborator: SK hynix
- ◇ Observed the behavior of a real CXL-based system on datacenter and AI workloads in the CXL-based platform
- ◇ Analyzed how the different promotion and demotion methods for CXL devices affected the performance of the workloads
- ◇ Presented performance modeling for datacenter workloads using different system factors (e.g., memory bandwidth, memory latency)
- ◇ Contributions: co-author, experiment, analysis, paper write-up

Accelerating Yinyang K-Means on Heterogenous Platform [LCTES'25]
Advisor: Prof. Yunho Oh, Korea University — Jun. 2024 - Sep. 2024

- ◇ Analyzed that CPU was underutilized while executing Yinyang K-Means clustering on embedded GPUs
- ◇ Observed warp divergence caused by checking boundary conditions for skip clustering degraded performance
- ◇ Developed a software technique to cooperate with CPUs to enhance performance on resource-constrained embedded GPUs
- ◇ Proposed an adaptive reordering for the condition check to balance the warps
- ◇ Contributions: co-author, idea, implementation, paper write-up

TLP Balancer: Predictive Thread Allocation for Multi-Tenant Inference in Embedded GPUs [ESL'24]
Advisor: Prof. Yunho Oh, Korea University — Mar. 2024 - Aug. 2024

- ◇ Observed fused kernels for multi-tenant AI workloads relied on the sub-optimal thread configuration
- ◇ Presented modeling to find the optimal thread configuration for fused AI kernels that balanced the warp-level computation
- ◇ Proposed a runtime system that dynamically fused AI kernels with the modeling
- ◇ Contributions: co-author, idea, paper write-up

VitBit: Enhancing Embedded GPU Performance for AI Workloads through Register Operand Packing [ICPP'24]
Advisor: Prof. Yunho Oh, Korea University — Sep. 2023 - May. 2024

- ◇ Observed under-utilization of floating CUDA cores or Tensor cores when processing integer-quantized AI workloads
- ◇ Proposed a software technique for simultaneous computation on all heterogeneous cores on GPU to support arbitrary integer formats
- ◇ Proposed a software-based packing policy to support simultaneous processing of packed integers
- ◇ Contributions: co-author, motivation study, idea, implementation, paper write-up

Salient Frequency-aware Exemplar Compression for Online Continual Learning [AAAI'25]
Supervisor: Dr. Suhyun Kim, Korea Institute of Science and Technology — Jan. 2023 - Nov. 2023

- ◇ Observed exemplar compression methods occupied limited GPU resources during online continual learning
- ◇ Proposed a computationally efficient compression algorithm using salient frequency
- ◇ Proposed a buffer management scheme to alleviate harmful effects from the compression artifacts remaining in the buffer
- ◇ Contributions: 1st author, motivation study, idea, implementation, paper write-up

Skills

C/C++, Python, Tensorflow, Pytorch, Git, Verilog, Shell script